

EX.NO:9 STRING OPERATIONS USING ARRAYLIST

Program:

```
import java.util.ArrayList;
import java.util.Scanner;

public class StringMenu {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<String> list = new ArrayList<>();
        int choice;
        do {
            System.out.println("\n--- STRING OPERATIONS ---");
            System.out.println("1. Append");
            System.out.println("2. Insert");
            System.out.println("3. Search");
            System.out.println("4. Display Starting Letter");
            System.out.println("5. Display All");
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Enter string: ");
                    list.add(sc.nextLine());
                    System.out.println("String added successfully.");
                    break;
                case 2:
                    System.out.print("Enter index: ");
                    int index = sc.nextInt();
                    sc.nextLine();
                    System.out.print("Enter string: ");
                    String value = sc.nextLine();
                    if (index >= 0 && index <= list.size()) {
                        list.add(index, value);
                        System.out.println("String inserted successfully.");
                    } else {
                        System.out.println("Invalid index.");
                    }
            }
        } while (choice != 6);
    }
}
```

```

        }
        break;
    case 3:
        System.out.print("Enter string to search: ");
        String search = sc.nextLine();
        if (list.contains(search))
            System.out.println("String found.");
        else
            System.out.println("String not found.");
        break;
    case 4:
        System.out.print("Enter starting letter: ");
        char ch = sc.nextLine().charAt(0);
        System.out.println("Matching strings:");
        for (String s : list) {
            if (Character.toLowerCase(s.charAt(0)) ==
Character.toLowerCase(ch)) {
                System.out.println(s);
            }
        }
        break;
    case 5:
        System.out.println("ArrayList: " + list);
        break;
    case 6:
        System.out.println("Program terminated.");
        break;
    default:
        System.out.println("Invalid choice.");
    }
} while (choice != 6);
sc.close();
}
}

```

Output:

--- STRING OPERATIONS ---

1. Append
2. Insert
3. Search
4. Display Starting Letter

5. Display All

6. Exit

Enter your choice: 1

Enter string: Apple

String added successfully.

--- STRING OPERATIONS ---

1. Append

2. Insert

3. Search

4. Display Starting Letter

5. Display All

6. Exit

Enter your choice: 1

Enter string: Banana

String added successfully.

--- STRING OPERATIONS ---

1. Append

2. Insert

3. Search

4. Display Starting Letter

5. Display All

6. Exit

Enter your choice: 2

Enter index: 1

Enter string: Orange

String inserted successfully.

--- STRING OPERATIONS ---

Enter your choice: 5

ArrayList: [Apple, Orange, Banana]

--- STRING OPERATIONS ---

Enter your choice: 3

Enter string to search: Banana

String found.

--- STRING OPERATIONS ---

Enter your choice: 4

Enter starting letter: A

Matching strings:
Apple

--- STRING OPERATIONS ---

Enter your choice: 6

Program terminated.